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Failure of Repeated Peer Separations to Induce Depression in Infant Rhesus Monkeys

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ABSTRACT

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Three rhesus monkeys (two female, one male) were separated from their mothers shortly after birth and were brought up together in a group cage. After 90 days together, they were subjected to eight repeated separations of 4–6 days during which they were housed in individual cages. Separations 7 and 8 were preceded by "vertical chamber" confinement of 3 and 8 days, respectively. Separation caused a marked increase in vocalizations. Although at times signs of depression (decreased locomotion, huddling) were apparent, they were neither frequent nor stable and did not seem to depend on either the length or the number of separations. It was concluded that social separation, although inducing clear signs of distress, does not invariably cause depression in infant rhesus monkeys.

Key words: animal model of depression, rhesus monkey, peer separation, distress vocalizations, motor activity.

INTRODUCTION

A major obstacle to research into the neurobiology of depression is the lack of adequate animal models. Most existing models, in particular those used in pharmacology for the detection of new drugs for the treatment of depression, were adopted because they had been found sensitive to clinically effective antidepressants. A weakness of these empirically based models is that they may only bear a chance relationship to depressive illness. Furthermore, they are unlikely to detect new compounds with mechanisms of action different from those already in use. There is thus a need for better models more closely related to clinically occurring depression.

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A promising candidate is the behavioral syndrome described in young rhesus monkeys separated either from their mothers or from their age mates (peers) or after confinement in "vertical chambers" [Mineka and Suomi, 1978]. The syndrome consists typically of two phases, first a high level of agitation and vocalization ("protest" phase) followed within about 48 hr by a marked reduction in vocalization and general activity ("despair" phase). Monkey depression not only bears a striking resemblance to "anaclitic" depression described in children separated from their mothers [Bowlby, 1977] but appears to respond to treatment with pharmacological agents such as imipramine [Suomi et al., 1978] and desipramine [Hrdina et al., 1979], which are known to alleviate depression in man.

Although the effects of mother-infant separation have been the most intensively studied [Mineka and Suomi, 1978], mother–infant separation has several practical disadvantages. First, the act of separation is in itself highly traumatic for both mother and infant requiring anesthesia of the mother or a struggle risking injury to the infant. More importantly, maturation of the infant with its increasing independence from the mother limits the number of separations that can be performed. For this reason the peer separation technique appears to be more promising. Suomi et al. [1970] have shown, for example, that four infant rhesus monkeys, separated from their mothers at birth and reared together, showed "protest-despair" reactions similar to those observed after maternal separation, every time they were separated from each other up to 20 times over a 7-month period without any signs of habituation. These observations suggested that it would be feasible to use the repeated peer separation procedure as a test system for the comparative study of different antidepressant drugs. The present paper describes attempts to induce signs of depression in infant rhesus monkeys reared together since birth, using a peer separation technique similar to that described by Suomi et al. [1970].

MATERIALS AND METHODS

Subjects

The subjects were three baby rhesus monkeys bred in our laboratory of different mothers but the same father and born, respectively, August 30 and September 18 and 20, 1981. They were removed from their mothers within 4 days after birth and placed into a stainless-steel cage ($70 \times 70 \times 60$ cm^3) in another laboratory with no other monkeys present. This meant that the first monkey was alone for 20 days before being joined by the other two monkeys. All time intervals mentioned below start from the day all three animals were together.

Rearing Procedures

The group cage contained a "surrogate mother" consisting of a piece of lambskin wrapped round a vertical column (40 cm high) and the floor was covered with multilayered absorbent crepe paper, which was changed daily. The temperature within the cage was maintained around 25°C by means of infrared heater. The surrogate mother, floor covering, and infrared heater were removed after 1 month. Subsequently, the floor consisted of a metal grid with a drop pan underneath; the ambient temperature of the laboratory was 21°C.

During the first month, the animals were fed on standard human baby formula prepared from powder and supplemented with cereals and soluble vitamins (Hydrosol:Roche, France). During the first 10 days after removal from the mother, each animal was hand-fed 5 times daily (8:30, 11:30, 14:30, 17:30, 21:30). During this time they were also taught to feed themselves from a bottle and teat attached to the cage. All three animals learned to self-feed within 10 days, whereafter they had free access to the formula, which was changed twice daily. During the second month the formula was replaced by ordinary milk supplemented as before. In addition fresh fruit and standard monkey pellets (U.A.R., Villemoison/Orge, France) were placed in the cage. From 2.5 months onward, the animals were maintained on a diet of standard monkey pellets with fresh fruit and they had free access to water enriched with soluble vitamins.

Separation Procedures

Beginning 3 months after placement in the group cage, the monkeys were subjected to a series of eight separations during which they were housed in individual cages ($70 \times 50 \times 30$ cm^3) in a small laboratory. The cages were constructed in white plastic with a stainless-steel wire grid front and were superimposed so that the animals could hear but not see each other. Each cage contained as a play object a small length of chain (15 cm) attached to the wire grid. During the separations the animals were left undisturbed apart from daily changing of the cage (9:00), two 15-min behavioral observations (11:30 and 14:00), and feeding (16:00). During the first six separations, the separation period lasted 4–6 days with intervals of 3–8 days between separations, during which the animals were housed together as described above. Before separations 7 and 8, in an attempt to intensify the behavioral phenomena, the animals were confined for 3 and 8 days, respectively, in stainless-steel vertical chambers similar to those used by Suomi and Harlow [1969] for inducing depression-like behavior in monkeys. After each confinement in the vertical chambers, the animals were placed in individual cages, as described above, and were observed for 4 days before being reunited in the group cage. During the seventh and eighth separations, the animals were thus alone for a total of 7 and 12 days, respectively.

During each separation, the animals were observed twice daily at 11:30 and 14:00. Observation sessions lasted 15 min, during which each subject was individually watched and its behavior scored for 5 min divided into ten 30-sec observation periods. During each 30-sec period, the presence (1) or absence (0) of the following behavioral categories was scored: locomotion, environmental exploration, stereotyped rocking, huddling, self-clasping, self-mouthing, and vocalization. These categories are defined according to Suomi et al. [1970, 1978]. The maximum possible daily score for each category, obtained by combining the scores from the morning and afternoon sessions, was thus 20. All observations were made by the same observer (R.D.P.). The observer always entered the experimental room 3 min before starting an observation and remained seated in the same position, visible to the animals, until the end of each 15-min session. During the last four separations the animals' behavior could also be monitored at any time without entering the room using a closed circuit television.

RESULTS

The only behavior to show clear-cut changes as a consequence of separation was the frequency of vocalization. The other parameters either showed no change (locomotion, environmental exploration, self-mouthing) or were hardly observed at all (stereotyped rocking, huddling, self-clasping). In order to simplify the presentation of the results, therefore, only the data for vocalizations and locomotion are presented. Frequency of locomotion was chosen because it was thought that changes in this measure, in particular a decrease in locomotion with increasing length of separation, should provide the best indicator of whether or not depression had been induced. All data were analyzed for statistical significance using a repeated measure analysis of variance.

The most marked effect of separation was a pronounced increase in the frequency of vocalizations over that observed when the animals were together. This is illustrated in Figure 1 for the first separation where the increase in vocalizations when the animals were separated compared with the preseparation baseline was highly significant (F treatment: 117.2, df = 1,14, $P < 0.001$). Similar patterns of group behavior with a virtual absence of vocalizations were observed throughout the study in the periods when the animals were together between separations (data not shown). Vocalizations were most frequent at the beginning of each separation and tended to decline as the separation continued (F days: 11.67, df = 3,64, $P < 0.001$). This same pattern was observed over eight repeated separations (Fig. 2). During separations 7 and 8 (Fig. 2), which were preceded by 3 and 8 days of vertical chamber

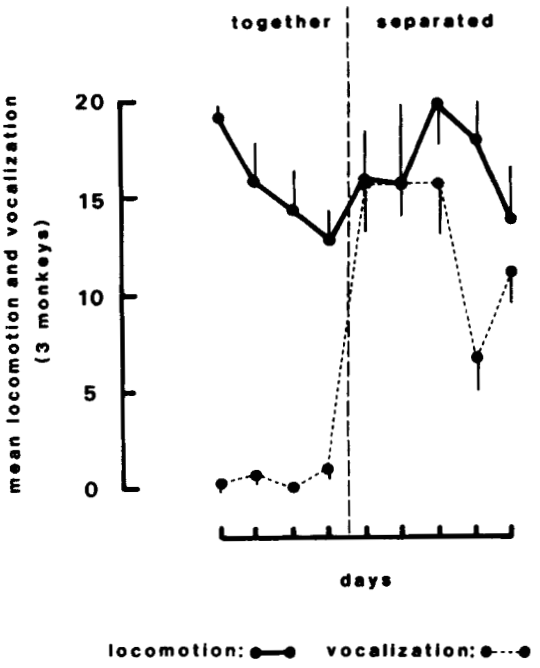


Fig. 1. Mean (± S.E.) vocalization and locomotion scores before and during separation 1.

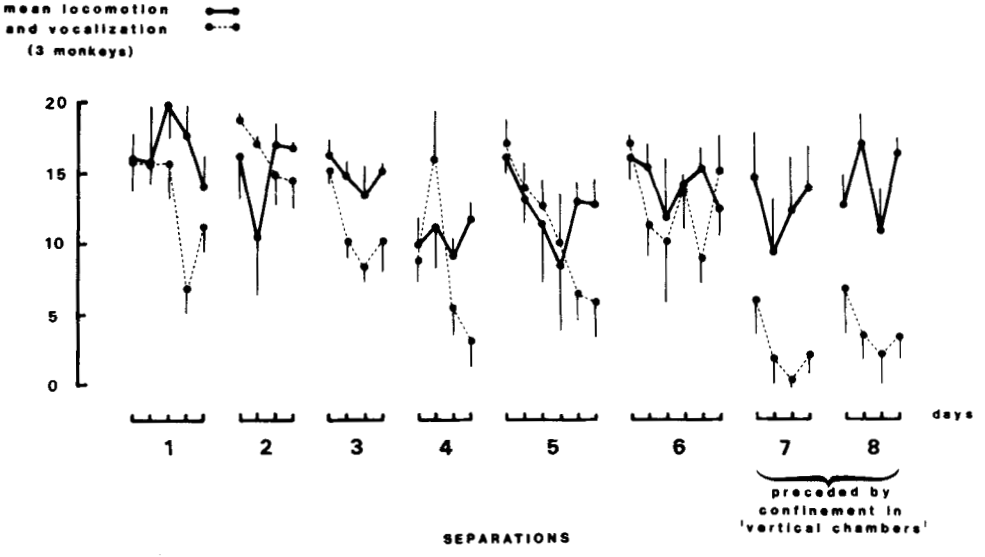


Fig. 2. Mean (± S.E.) vocalization and locomotion scores during eight separations.

confinement, respectively, the initial frequency of vocalization was lower than during the previous six separations (F separation: 22.25, df = 7,64, $P < 0.001$), and it rapidly declined to levels similar to those occurring when the animals were together (see Fig. 1).

In contrast there were no marked changes in the frequency of locomotion either when compared to preseparation values (Fig. 1: F treatment: 1.54, df = 1,14, $P > 0.05$) or within

each individual separation (Fig. 2: F days: < 1). Although the animals appeared more agitated at the beginning of each separation, this was not reflected in their locomotion scores and, in particular, there was no systematic decrease in locomotion within each separation as would have been expected had the animals become depressed. Even after 3 or 8 days of vertical chamber confinement (separations 7 and 8), the locomotion scores remained essentially similar to those obtained during the previous six separations.

There were, nonetheless, occasions when one or another animal would remain sitting motionless for extended periods in a hunched position (huddling) with a forlorn expression on its face sometimes emitting plaintive cooing sounds. Any movements occurring during these phases appeared to be very slow; the animals frequently appeared to be half asleep with their eyes almost closed, but they remained totally responsive to the slightest environmental stimulus. These phases of apparent depression were neither frequent nor stable. They did not increase as a function of either the length of each separation or the number of separations and were totally absent after vertical chamber confinement of 3 or 8 days (separations 7 and 8). At no time were more marked behavioral signs observed, such as prostration, self-clasping, stereotyped rocking, refusal to eat, or indifference to environmental stimuli, which might have suggested severe emotional disturbance.

The results of these standardized observations were confirmed by more informal observations made at various times of the day by means of the closed circuit television (separations 5 to 8), suggesting that the failure to observe depression was not due to inappropriate behavior sampling procedures or to the presence of the observer.

DISCUSSION

These experiments failed, therefore, to induce convincing signs of depression in three baby monkeys using a repeated peer separation technique similar to that described by Suomi et al. [1970]. Although clear signs of "protest" in the form of marked vocalizations were observed at the beginning of each separation, in no case was there any systematic decrease in locomotion as each separation continued, which, together with self-clasping, rocking, or huddling, might have reflected "despair." One perhaps important difference between the present procedure and that described by Suomi et al. [1970] was that the infants were raised together with a furry mother surrogate during the first month of life, whereas the previous study mentions only the presence of cloth towels. Access to a furry surrogate to which the animals can cling as if to a real mother has a very calming influence on infant monkeys and may have diminished the amount of time they would otherwise have spent clinging to each other at a perhaps critical stage in the formation of affectional bonds [Mineka and Suomi, 1978]. It should be noted, nonetheless, that on separation from each other the animals showed clear signs of distress, suggesting that being separated was still an aversive experience. The possibility remains that the three half-sibling monkeys studied were constitutionally not susceptible to depression. There exists considerable variability between different monkey species in the facility with which depression-like behavior can be induced [Mineka and Suomi, 1978] and even in the rhesus monkey, individual differences are common. For example, in one study of the effects of separation in 20 rhesus monkey mother–infant pairs, clear signs of depression (self-clasping, huddling, and reduced locomotion) were observed in only 3 out of the 20 infants tested [Lewis et al., 1976], whereas in another study with 7 mother–infant pairs, a clear and predominant despair response was not seen at all [Gunnar et al., 1981].

It is possible that longer periods of confinement in vertical chambers might have eventually caused the animals to become depressed [Suomi and Harlow, 1972]. However, longer periods would severely limit the utility of this model for the comparative study of antidepressant compounds, and for this reason the experiments were abandoned. In general, our results suggest that peer separation, like mother–infant separation does not invariably induce behavioral changes that can be unequivocally described as depression in young rhesus monkeys.

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